2.73 3D Matrices: Diagonalisation

Worked Solutions

1 a)

$$\mathbf{A}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix},\quad \mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}1-\lambda & 0 & -4\\ 0 & 2-\lambda & -5\\ 0 & 0 & -3-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=(1-\lambda)(2-\lambda)(-3-\lambda)-0+0$$

$$(1-\lambda)(2-\lambda)(-3-\lambda)=0\Rightarrow\lambda_1=-3,\ \lambda_2=1 \text{ or } \lambda_3=2,\quad \therefore\lambda_3=2.$$

1 b)

$$\lambda_3=2:\quad \mathbf{A}-2\mathbf{I}=\begin{pmatrix}-1 & 0 & -4\\ 0 & 0 & -5\\ 0 & 0 & -5\end{pmatrix},\quad (\mathbf{A}-2\mathbf{I})\begin{pmatrix}x\\y\\z\end{pmatrix}=\mathbf{0}$$

$$-x-4z=0\Rightarrow x=-4z,\qquad -5z=0$$

$$z=0 \text{ so that } x=z=0,\qquad \therefore\mathbf{v}_3=\begin{pmatrix}0\\1\\0\end{pmatrix}$$

$$\mathbf{P}=\begin{pmatrix}1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 0 & 0\end{pmatrix},\quad \det\mathbf{P}=\begin{vmatrix}1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 0 & 0\end{vmatrix}=-1(-1)+0-0=1$$

$$\text{Cofactor matrix of }\mathbf{P}=\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & -1 & -1\end{pmatrix},\quad \therefore\mathbf{P}^{-1}=\begin{pmatrix}0 & 0 & 1\\ 1 & 0 & -1\\ 0 & 1 & -1\end{pmatrix}.$$

1 c)

$$\mathbf{D}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}=\begin{pmatrix}0 & 0 & 1\\ 1 & 0 & -1\\ 0 & 1 & -1\end{pmatrix}\begin{pmatrix}1 & 0 & -4\\ 0 & 2 & -5\\ 0 & 0 & -3\end{pmatrix}\begin{pmatrix}1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 0 & 0\end{pmatrix}$$

$$=\begin{pmatrix}0 & 0 & 1\\ 1 & 0 & -1\\ 0 & 1 & -1\end{pmatrix}\begin{pmatrix}-3 & 1 & 0\\ -3 & 0 & 2\\ -3 & 0 & 0\end{pmatrix}=\begin{pmatrix}-3 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 2\end{pmatrix}.$$

2 a)

$$\mathbf{A}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix},\quad \mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}5-\lambda & 1 & -5\\ 2 & 3-\lambda & -4\\ 2 & 1 & -2-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I}) = \begin{vmatrix} 5-\lambda & 1 & -5 \\ 2 & 3-\lambda & -4 \\ 2 & 1 & -2-\lambda \end{vmatrix} = \begin{vmatrix} 5-\lambda & 1 & -5 \\ 0 & 2-\lambda & -2+\lambda \\ 2 & 1 & -2-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} 5-\lambda & 1 & -4 \\ 0 & 2-\lambda & 0 \\ 2 & 1 & -1-\lambda \end{vmatrix} = \begin{vmatrix} 3-\lambda & 0 & -3+\lambda \\ 0 & 2-\lambda & 0 \\ 2 & 1 & -1-\lambda \end{vmatrix}$$

$$= (3-\lambda)((2-\lambda)(-1-\lambda)) - 2(-3+\lambda)(2-\lambda)$$

$$= (3-\lambda)(2-\lambda)(-1-\lambda+2) = (3-\lambda)(2-\lambda)(1-\lambda).$$

$$(3-\lambda)(2-\lambda)(1-\lambda) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3.$$

$$\lambda_1 = 1: \quad \mathbf{A}-\mathbf{I} = \begin{pmatrix} 4 & 1 & -5 \\ 2 & 2 & -4 \\ 2 & 1 & -3 \end{pmatrix}, \quad (\mathbf{A}-\mathbf{I})\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$4x + y - 5z = 0\ (1), \quad 2x + 2y - 4z = 0\ (2), \quad 2x + y - 3z = 0\ (3)$$

$$(1)-(2): \quad 2x - 2z = 0 \Rightarrow x = z, \quad \text{sub into (3): } y - x = 0 \Rightarrow y = x, \quad \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

$$\lambda_2 = 2: \quad \mathbf{A}-2\mathbf{I} = \begin{pmatrix} 3 & 1 & -5 \\ 2 & 1 & -4 \\ 2 & 1 & -4 \end{pmatrix}, \quad (\mathbf{A}-2\mathbf{I})\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$3x + y - 5z = 0\ (1), \quad 2x + y - 4z = 0\ (2), \quad 2x + y - 4z = 0\ (3)$$

$$(1)-(2): \quad x - z = 0 \Rightarrow x = z, \quad \text{sub into (2): } y - 2z = 0 \Rightarrow y = 2z, \quad \therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$\lambda_3 = 3: \quad \mathbf{A}-3\mathbf{I} = \begin{pmatrix} 2 & 1 & -5 \\ 2 & 0 & -4 \\ 2 & 1 & -5 \end{pmatrix}, \quad (\mathbf{A}-3\mathbf{I})\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$2x + y - 5z = 0\ (1), \quad 2x - 4z = 0\ (2), \quad 2x + y - 5z = 0\ (3)$$

$$(2): \quad x - 2z = 0 \Rightarrow x = 2z, \quad \text{sub into (1): } y - z = 0 \Rightarrow y = z, \quad \therefore \mathbf{v}_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} -1 & -1 & 3 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix}, \quad \mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

$$= \begin{pmatrix} -1 & -1 & 3 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} 5 & 1 & -5 \\ 2 & 3 & -4 \\ 2 & 1 & -2 \end{pmatrix}\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\therefore \mathbf{P} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

2 b)

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{A}-\lambda\mathbf{I} = \begin{pmatrix} 1-\lambda & 2 & 1 \\ 2 & 3-\lambda & 1 \\ 1 & 4 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I}) = \begin{vmatrix} 1-\lambda & 2 & 1 \\ 2 & 3-\lambda & 1 \\ 1 & 4 & 3-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & 2 & 1 \\ 2 & 3-\lambda & 1 \\ -1+2\lambda & 0 & 1-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} 1-\lambda & 2 & 1 \\ 1+\lambda & 1-\lambda & 0 \\ -1+2\lambda & 0 & 1-\lambda \end{vmatrix}$$

$$= 1(0-(1-\lambda)(-1+2\lambda)) - 0 + (1-\lambda)((1-\lambda)(1-\lambda) - 2(1+\lambda))$$

$$= -(1-\lambda)(-1+2\lambda) + (1-\lambda)(\lambda^2 - 4\lambda - 1)$$

$$= (1-\lambda)(1-2\lambda+\lambda^2-4\lambda-1) = (1-\lambda)(\lambda^2-6\lambda) = \lambda(1-\lambda)(\lambda-6).$$

$$\lambda(1-\lambda)(\lambda-6) = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 6.$$

$$\lambda_1 = 0: \quad \mathbf{A} - 0\mathbf{I} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 4 & 3 \end{pmatrix}, \quad x + 2y + z = 0\ (1),\ 2x + 3y + z = 0\ (2),\ x + 4y + 3z = 0\ (3)$$

$$(3)-(1): \quad 2y + 2z = 0 \Rightarrow z = -y, \quad \text{sub into (1): } x + y = 0 \Rightarrow x = -y, \quad \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

$$\lambda_2 = 1: \quad \mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 4 & 2 \end{pmatrix}, \quad 2y + z = 0\ (1),\ 2x + 2y + z = 0\ (2),\ x + 4y + 2z = 0\ (3)$$

$$\text{From (1): } 2y + z = 0 \Rightarrow -z = 2y, \quad \text{sub into (2): } 2x = 0 \Rightarrow x = 0, \quad \therefore \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}.$$

$$\lambda_3 = 6: \quad \mathbf{A} - 6\mathbf{I} = \begin{pmatrix} -5 & 2 & 1 \\ 2 & -3 & 1 \\ 1 & 4 & -3 \end{pmatrix}, \quad -5x + 2y + z = 0\ (1),\ 2x - 3y + z = 0\ (2),\ x + 4y - 3z = 0\ (3$$

$$(1)-(2): \quad -7x + 5y + z = 0 \Rightarrow 7x = 5y \Rightarrow y = \tfrac{7}{5}x, \quad \Rightarrow \tfrac{7}{5}x = y$$

$$\text{sub into (3): } \tfrac{33}{5}x - 3z = 0 \Rightarrow \tfrac{11}{5}x = z, \quad \therefore \mathbf{v}_3 = \begin{pmatrix} 5 \\ 7 \\ 11 \end{pmatrix}.$$

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 5 \\ -1 & 1 & 7 \\ 1 & -2 & 11 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{30}\begin{pmatrix} 25 & -10 & -5 \\ 18 & 6 & -12 \\ 1 & 2 & 1 \end{pmatrix}, \quad \mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

$$= \frac{1}{30}\begin{pmatrix} 25 & -10 & -5 \\ 18 & 6 & -12 \\ 1 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 4 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 5 \\ -1 & 1 & 7 \\ 1 & -2 & 11 \end{pmatrix}$$

$$= \frac{1}{30}\begin{pmatrix} 25 & -10 & -5 \\ 18 & 6 & -12 \\ 1 & 2 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 30 \\ 0 & 1 & 42 \\ 0 & -2 & 66 \end{pmatrix} = \frac{1}{30}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 30 & 0 \\ 0 & 0 & 180 \end{pmatrix}$$

$$\therefore \mathbf{P} = \begin{pmatrix} 1 & 0 & 5 \\ -1 & 1 & 7 \\ 1 & -2 & 11 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}.$$

3 a)

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (2-\lambda)(2-\lambda)(3-\lambda) = 0 \Rightarrow \lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 3.$$

$$\lambda_1 = 2: \quad \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad y = 0\ (1),\ z = 0\ (3), \quad \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

$$\lambda_2 = \lambda_3 = 3: \quad \mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad -x + y = 0\ (1),\ -y = 0\ (2)$$

$$\therefore x = y = 0, \quad \mathbf{v}_2 = \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \therefore \lambda_1 = 2, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \lambda_2 = \lambda_3 = 3, \mathbf{v}_2 = \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

3 b)

We need 3 independent eigenvectors for a 3×3 matrix to be diagonalisable but we have only 2.

4 a)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^6 = \begin{pmatrix} 2^6 & 0 & 0 \\ 0 & (-1)^6 & 0 \\ 0 & 0 & 1^6 \end{pmatrix} = \begin{pmatrix} 64 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

4 b)

$$\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix}^9 = \begin{pmatrix} (\frac{1}{2})^9 & 0 & 0 \\ 0 & (-1)^9 & 0 \\ 0 & 0 & (-2)^9 \end{pmatrix} = \begin{pmatrix} \frac{1}{512} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -512 \end{pmatrix}.$$

4 c)

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^k = \begin{pmatrix} a^k & 0 & 0 \\ 0 & b^k & 0 \\ 0 & 0 & c^k \end{pmatrix}.$$

5)

$$\mathbf{P} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad \mathbf{A} = \mathbf{PDP}^{-1}, \quad \mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1}$$

$$\mathbf{P}^{-1} = \frac{1}{6}\begin{pmatrix} 4 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 3 & 1 \end{pmatrix}$$

$$\mathbf{A}^4 = \frac{1}{6}\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}^4\begin{pmatrix} 4 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 3 & 1 \end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 81 \end{pmatrix}\begin{pmatrix} 4 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 3 & 1 \end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix} 6 & 0 & 0 \\ -80 & 246 & 80 \\ -240 & 720 & 246 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ -40 & 123 & 40 \\ -120 & 360 & 123 \end{pmatrix}.$$

6)

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{M}^{10} = \mathbf{P}\mathbf{D}^{10}\mathbf{P}^{-1}, \quad \mathbf{P}^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 2 & 1 \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$$

$$\mathbf{M}^{10} = \frac{1}{5}\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} (-5)^{10} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -3 & 2 & 1 \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$$

$$= \frac{1}{5}\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} -3 \times 5^{10} & 2 \times 5^{10} & 5^{10} \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$$

$$= \frac{1}{5}\begin{pmatrix} 5 & 0 & 0 \\ -6 \times 5^{10} + 6 & 4 \times 5^{10} + 1 & 2 \times 5^{10} - 2 \\ -3 \times 5^{10} + 3 & 2 \times 5^{10} - 2 & 5^{10} + 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ -6 \times 5^{9} + \frac{6}{5} & 4 \times 5^{9} + \frac{1}{5} & 2 \times 5^{9} - \frac{2}{5} \\ -3 \times 5^{9} + \frac{3}{5} & 2 \times 5^{9} - \frac{2}{5} & 5^{9} + \frac{4}{5} \end{pmatrix}.$$

7 a)

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{D} = \mathbf{P}^{-1}\mathbf{M}\mathbf{P} \Rightarrow \mathbf{M} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1} \therefore \mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$$

$$\mathbf{P}^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 1 & 1 \\ 2 & 0 & -1 \end{pmatrix}$$

$$\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}\begin{pmatrix} (\frac{1}{2})^n & 0 & 0 \\ 0 & (\frac{1}{3})^n & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 & 1 \\ -2 & 1 & 1 \\ 2 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}\begin{pmatrix} -(\frac{1}{2})^n & 0 & (\frac{1}{2})^n \\ -2(\frac{1}{3})^n & (\frac{1}{3})^n & (\frac{1}{3})^n \\ 2 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 - (\frac{1}{2})^n & 0 & (\frac{1}{2})^n - 1 \\ 2 - 2(\frac{1}{3})^n & (\frac{1}{3})^n & (\frac{1}{3})^n - 1 \\ 2 - 2(\frac{1}{2})^n & 0 & 2(\frac{1}{2})^n - 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 - \frac{1}{2^n} & 0 & \frac{1}{2^n} - 1 \\ 2 - \frac{2}{3^n} & \frac{1}{3^n} & \frac{1}{3^n} - 1 \\ 2 - \frac{2}{2^n} & 0 & \frac{2}{2^n} - 1 \end{pmatrix}.$$

7 b)

$$\mathbf{L} = \lim_{n\to\infty} \mathbf{M}^n = \lim_{n\to\infty} \begin{pmatrix} 2-\frac{1}{2^n} & 0 & \frac{1}{2^n}-1 \\ 2-\frac{2}{3^n} & \frac{1}{3^n} & \frac{1}{3^n}-1 \\ 2-\frac{2}{2^n} & 0 & \frac{2}{2^n}-1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & -1 \\ 2 & 0 & -1 \\ 2 & 0 & -1 \end{pmatrix}.$$

8 a)

$$\mathbf{P} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}, \quad \mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}, \quad \mathbf{P}^{-1} = \frac{1}{6}\begin{pmatrix} 3 & 3 & -3 \\ 3 & -1 & -1 \\ -3 & -1 & 5 \end{pmatrix}$$

$$\begin{aligned} \mathbf{M}^n &= \frac{1}{6}\begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & (\frac{3}{4})^n & 0 \\ 0 & 0 & (\frac{1}{5})^n \end{pmatrix}\begin{pmatrix} 3 & 3 & -3 \\ 3 & -1 & -1 \\ -3 & -1 & 5 \end{pmatrix} \\ &= \frac{1}{6}\begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} 3 & 3 & -3 \\ 3(\frac{3}{4})^n & -(\frac{3}{4})^n & -(\frac{3}{4})^n \\ -3(\frac{1}{5})^n & -(\frac{1}{5})^n & 5(\frac{1}{5})^n \end{pmatrix} \\ &= \frac{1}{6}\begin{pmatrix} 3+6(\frac{3}{4})^n-\frac{3}{5^n} & 3-2(\frac{3}{4})^n-\frac{1}{5^n} & -3-2(\frac{3}{4})^n-\frac{3}{5^n} \\ 6-3(\frac{3}{4})^n-\frac{3}{5^n} & 6+(\frac{3}{4})^n-\frac{1}{5^n} & -6+(\frac{3}{4})^n-\frac{3}{5^n} \\ 3+3(\frac{3}{4})^n-\frac{6}{5^n} & 3-(\frac{3}{4})^n-\frac{2}{5^n} & -3-(\frac{3}{4})^n-\frac{6}{5^n} \end{pmatrix}. \end{aligned}$$

8 b)

$$\mathbf{L} = \lim_{n\to\infty} \mathbf{M}^n = \frac{1}{6}\begin{pmatrix} 3 & 3 & -3 \\ 6 & 6 & -6 \\ 3 & 3 & -3 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}.$$

9)

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{A}-\lambda\mathbf{I} = \begin{pmatrix} k-\lambda & 0 & k \\ 0 & k-\lambda & 0 \\ k & 0 & k-\lambda \end{pmatrix}$$

$$\begin{aligned} \det(\mathbf{A}-\lambda\mathbf{I}) &= (k-\lambda)^3 - k^2(k-\lambda) = (k-\lambda)((k-\lambda)^2-k^2) \\ &= (k-\lambda)(k^2-2k\lambda+\lambda^2-k^2) = \lambda(k-\lambda)(-2k+\lambda) \end{aligned}$$

$$\lambda(k-\lambda)(\lambda-2k) = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = k, \lambda_3 = 2k.$$

$$\lambda_1 = 0: \quad \mathbf{A}-0\mathbf{I} = \begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix}, \quad kx+kz = 0\ (1),\ ky = 0\ (2)$$

$$\therefore z = -x,\ y = 0, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

$$\lambda_2 = k: \quad \mathbf{A}-k\mathbf{I} = \begin{pmatrix} 0 & 0 & k \\ 0 & 0 & 0 \\ k & 0 & 0 \end{pmatrix}, \quad kz = 0\ (1),\ kx = 0\ (2)$$

$$\therefore x = 0,\ z = 0, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

$$\lambda_3 = 2k: \quad \mathbf{A} - k\mathbf{I} = \begin{pmatrix} -k & 0 & k \\ 0 & -k & 0 \\ k & 0 & -k \end{pmatrix}, \quad -kx + kz = 0\ (1),\ -ky = 0\ (2)$$

$$\Rightarrow x = z,\ y = 0, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 2k \end{pmatrix}$$

$$\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$$

$$= \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & k^n & 0 \\ 0 & 0 & 2^n k^n \end{pmatrix}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= k^n \begin{pmatrix} 2^{n-1} & 0 & 2^{n-1} \\ 0 & 1 & 0 \\ 2^{n-1} & 0 & 2^{n-1} \end{pmatrix} \text{ as req.}$$

10 a)

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} a - \frac{b}{2} - \lambda & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b - \lambda & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} - \lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} a - \frac{b}{2} - \lambda & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b - \lambda & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} - \lambda \end{vmatrix} = \begin{vmatrix} a - \lambda & \frac{b}{2} & -a + \frac{b}{2} \\ a - \lambda & b - \lambda & -a + b \\ 0 & \frac{b}{2} & \frac{b}{2} - \lambda \end{vmatrix}$$

$$= \begin{vmatrix} a - \lambda & \frac{b}{2} & -a + \frac{b}{2} \\ 0 & \frac{b}{2} - \lambda & \frac{b}{2} \\ 0 & \frac{b}{2} & \frac{b}{2} - \lambda \end{vmatrix} = \begin{vmatrix} a - \lambda & \frac{b}{2} & -a + \frac{b}{2} \\ 0 & -\lambda & \lambda \\ 0 & \frac{b}{2} & \frac{b}{2} - \lambda \end{vmatrix}$$

$$= \begin{vmatrix} a - \lambda & \frac{b}{2} & -a \\ 0 & -\lambda & 2\lambda \\ 0 & \frac{b}{2} & -\lambda \end{vmatrix} = (a - \lambda)(\lambda^2 - b\lambda)$$

$$= \lambda(a - \lambda)(\lambda - b), \quad \therefore \lambda_1 = 0, \lambda_2 = a, \lambda_3 = b.$$

10 b)

$$\lambda_1 = 0: \quad \mathbf{A} - 0\mathbf{I} = \begin{pmatrix} a - \frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix}, \quad (\mathbf{A} - 0\mathbf{I}) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\left(a - \frac{b}{2}\right)x + \frac{b}{2}y + \left(-a + \frac{b}{2}\right)z = 0 \ (1), \quad (a-b)x + by + (-a+b)z = 0 \ (2), \quad -\frac{b}{2}x + \frac{b}{2}y + \frac{b}{2}z = 0 \ (3)$$

$$(1) - (3): \quad ax - az = 0 \Rightarrow x = z, \quad \text{sub into (2): } by = 0 \Rightarrow y = 0, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

$$\lambda_2 = a: \quad \mathbf{A} - a\mathbf{I} = \begin{pmatrix} -\frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & -a + b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \end{pmatrix}, \quad (\mathbf{A} - a\mathbf{I}) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$-\frac{b}{2}x + \frac{b}{2}y + \left(-a + \frac{b}{2}\right)z = 0 \ (1), \quad (a-b)x + (-a+b)y + (-a+b)z = 0 \ (2), \quad -\frac{b}{2}x + \frac{b}{2}y + \left(-a + \frac{b}{2}\right)z = 0 \ (3)$$

$$(1) - 2(3): \quad ax - ay + az = 0 \Rightarrow x - y + z = 0 \Rightarrow y = x + z, \quad \text{sub into (2): } (-2a + 2b)z = 0 \Rightarrow z = 0, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

$$\lambda_3 = a: \quad \mathbf{A} - b\mathbf{I} = \begin{pmatrix} a - \frac{3b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & 0 & -a + b \\ -\frac{b}{2} & \frac{b}{2} & -\frac{b}{2} \end{pmatrix}, \quad (\mathbf{A} - b\mathbf{I}) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\left(a - \frac{3b}{2}\right)x + \frac{b}{2}y + \left(-a + \frac{b}{2}\right)z = 0 \ (1), \quad (a-b)x + (-a+b)z = 0 \ (2), \quad -\frac{b}{2}x + \frac{b}{2}y - \frac{b}{2}z = 0 \ (3)$$

$$(2): \quad x = z, \quad \text{sub into (3): } \frac{1}{2}y - z = 0 \Rightarrow y = 2z, \quad \text{so } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}, \quad \mathbf{D} = \mathbf{P}^{-1}\mathbf{M}\mathbf{P} \Rightarrow \mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$$

$$\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & -2 \\ -1 & 1 & 1 \end{pmatrix}$$

$$\mathbf{M}^n = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & a^n & 0 \\ 0 & 0 & b^n \end{pmatrix}\begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & -2 \\ -1 & 1 & 1 \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 2a^n & 0 & -2a^n \\ -b^n & b^n & b^n \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} 2a^n - b^n & b^n & b^n - 2a^n \\ 2a^n - 2b^n & 2b^n & 2b^n - 2a^n \\ -b^n & b^n & b^n \end{pmatrix} \text{ as req.}$$